Comparing input root files

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Input Files Used

Source 1: Hugo (All-in-one)

/root/github/e906-development/ROOTFiles/Hugo/roadset57_70_R008_2111v42_tmp_noPhys.root

Source 2: Harsha (Target/Roadset Split)*

/root/github/e906-development/ROOTFiles/Harsha/merged_RS[RS]_[Target]_[Pos]_[Num]_FinalData.root

Source 3: Abi (Mixed Events - RS67 Only)

/root/github/e906-development/ROOTFiles/MixedEvents/merged_RS67_3089LH2.root
/root/github/e906-development/ROOTFiles/MixedEvents/merged_RS67_3089LD2.root
/root/github/e906-development/ROOTFiles/MixedEvents/merged_RS67_3089flask.root

*RS = [57, 59, 62, 70], Target = [LH2, LD2, Flask]

Event Selection (Dimuon Cuts)

- **Dynamic Offset (bo):** 1.6 if $runID \geq 11000$, else 0.4
- **Vertex & Geometry:** $|dx| < 0.25$, $|dy - bo| < 0.22$, $-280 < dz < -5$
- **Kinematics:** $|dpx| < 1.8$, $|dpy| < 2.0$, $dpx^2 + dpy^2 < 5$, $38 < dpz < 116$
- **Mass & Target Setup:** $4.2 < \text{mass} < 8.8$, $dx^2 + (dy - bo)^2 < 0.06$
- **Physics Variables:** $-0.1 < xF < 0.95$, $0.05 < xT \leq 0.58$, $|\cos\theta| < 0.5$
- **Quality:** $|\text{trackSeparation}| < 270$, $\chi^2_{\text{dimuon}} < 18$

Event Selection (Single Track $i \in \{1, 2\}$ Cuts)

- **Target & Station 1:** $\chi^2_{i,\text{target}} < 15,\ 9 < pz_{i,\text{st1}} < 75,\ nHits_i > 13$
- **Positions:** $x^2_{i,t} + (y_{i,t} - bo)^2 < 320,\ 16 < x^2_{i,d} + (y_{i,d} - bo)^2 < 1100$
- χ^2 **Comparisons:** $\chi^2_{i,\text{target}} < 1.5\chi^2_{i,\text{upstream}},\ \chi^2_{i,\text{target}} < 1.5\chi^2_{i,\text{dump}}$
- **Vertex Z & Fit Quality:** $-320 < z_{i,v} < -5,\ \chi^2_i/(nHits_i - 5) < 12$
- **Station 1 vs Station 3 Tracking:**
  - $y_{i,\text{st1}}/y_{i,\text{st3}} < 1,\ y_{i,\text{st1}} \cdot y_{i,\text{st3}} > 0$
  - $||px_{i,\text{st1}} - px_{i,\text{st3}}| - 0.416| < 0.008$
  - $|py_{i,\text{st1}} - py_{i,\text{st3}}| < 0.008,\ |pz_{i,\text{st1}} - pz_{i,\text{st3}}| < 0.08$
  - $|py_{i,\text{st1}}| > 0.02$

Event Selection (Combined Tracks & Occupancy)

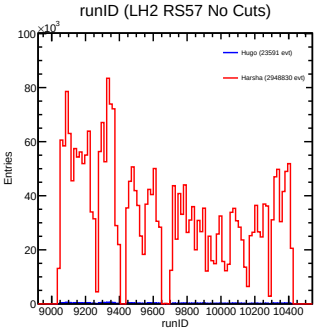
- **Combined Track Cuts:**
  - $|\chi^2_{1,\text{target}} + \chi^2_{2,\text{target}} - \chi^2_{\text{dimuon}}| < 2$
  - $y_{1,\text{st3}} \cdot y_{2,\text{st3}} < 0,\ nHits_1 + nHits_2 > 29,\ nHits1St1 + nHits2St1 > 8$
  - $|x_{1,\text{st1}} + x_{2,\text{st1}}| < 42$
- **Detector Occupancy Cuts:**
  - $D1 < 400,\ D2 < 400,\ D3 < 400, D1 + D2 + D3 < 1000$

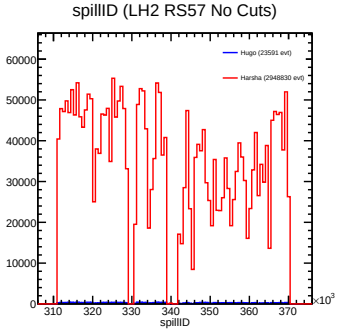
Roadsets and Run Ranges

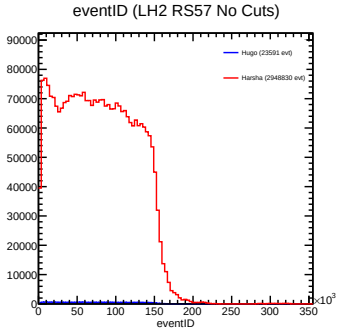
Roadset	runID Range
RS57	8912 - 10420
RS59	10421 - 10912
RS62	11075 - 12438
RS67	12525 - 15789
RS70	15793 - 16076

Plots Without Event Selection

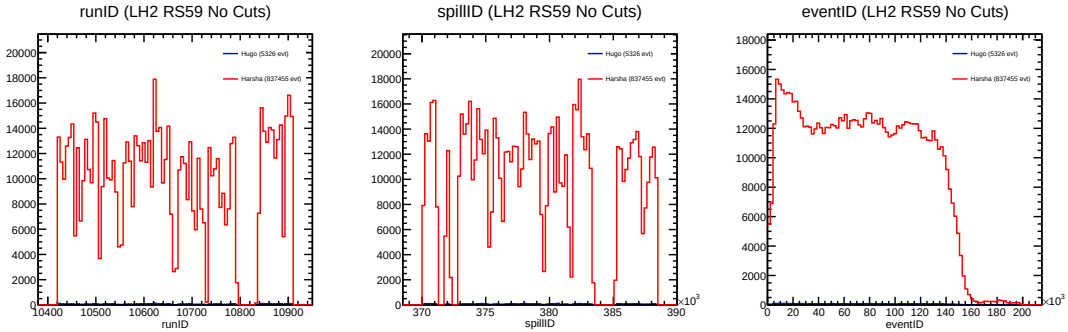
No Selection: LH2 Roadset 57



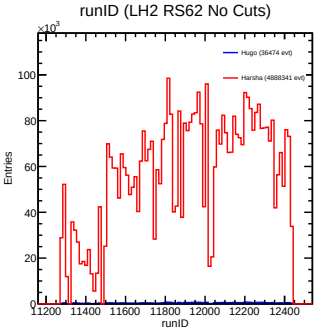


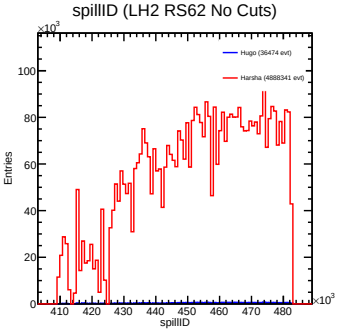


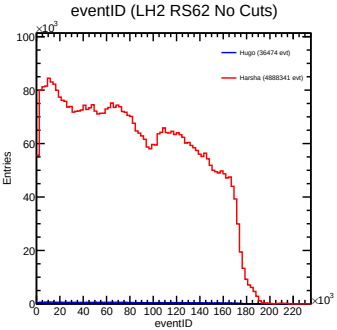
No Selection: LH2 Roadset 59



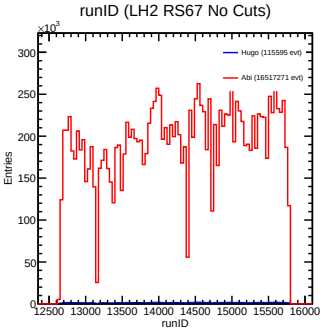
No Selection: LH2 Roadset 62

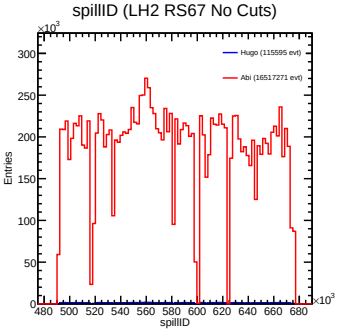


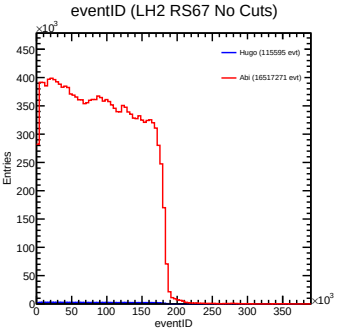




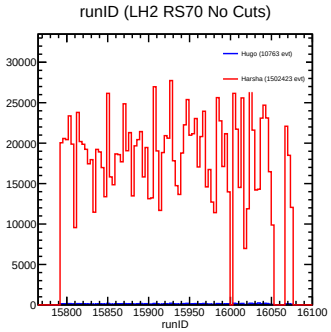
No Selection: LH2 Roadset 67

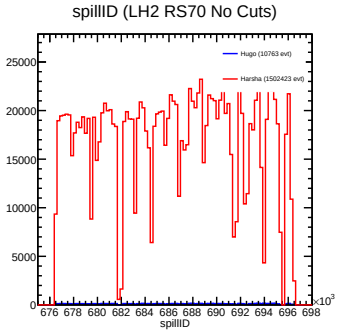


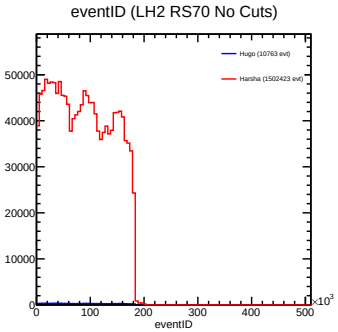




No Selection: LH2 Roadset 70

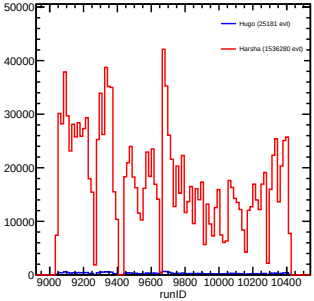




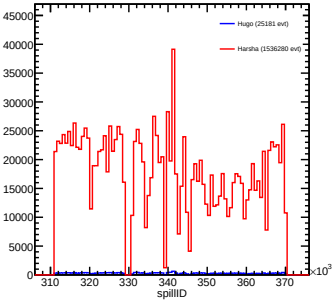


No Selection: LD2 Roadset 57

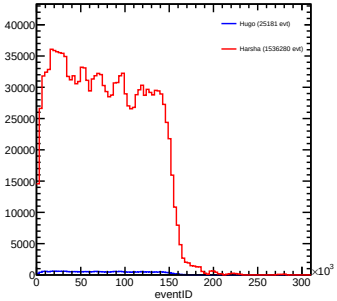
runID (LD2 RS57 No Cuts)



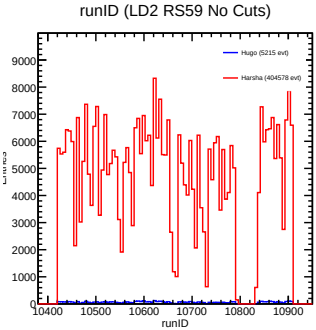
spillID (LD2 RS57 No Cuts)

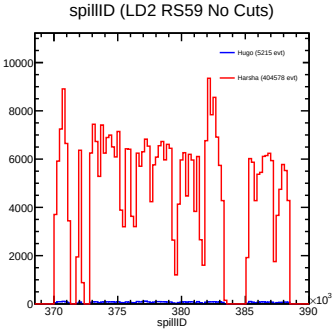


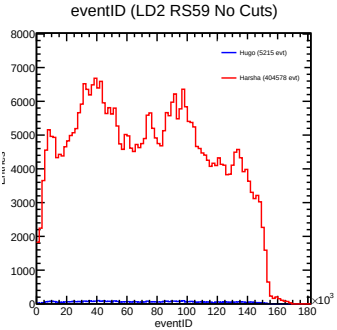
eventID (LD2 RS57 No Cuts)



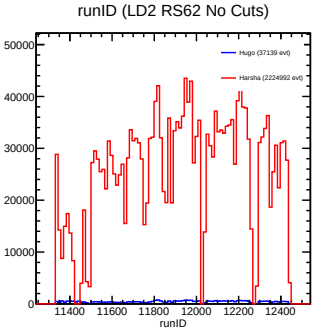
No Selection: LD2 Roadset 59

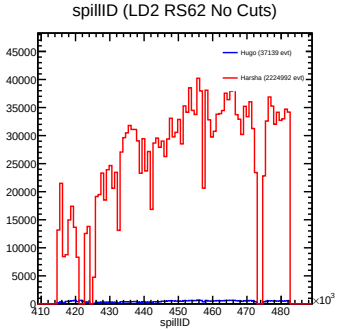


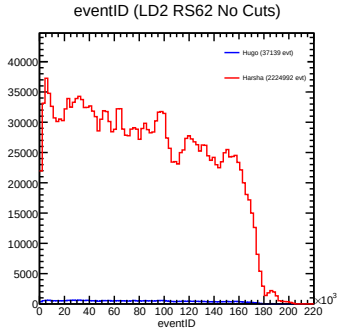




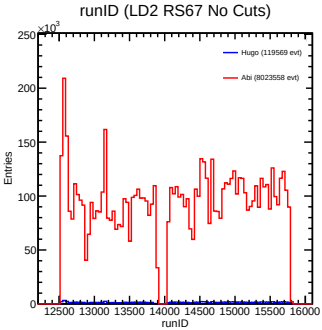
No Selection: LD2 Roadset 62

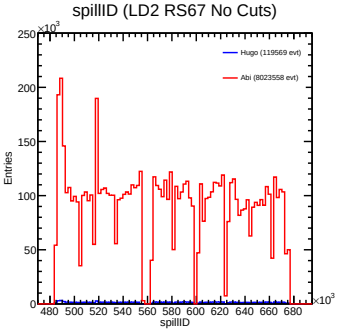


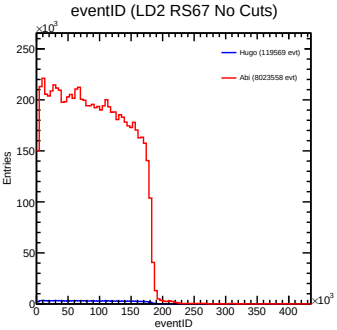




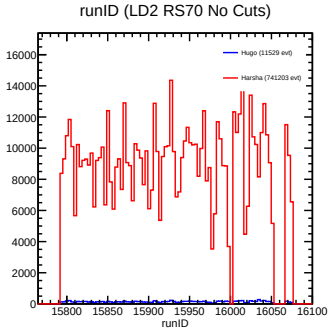
No Selection: LD2 Roadset 67

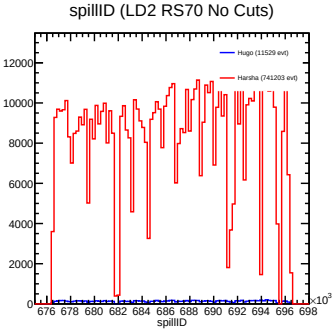


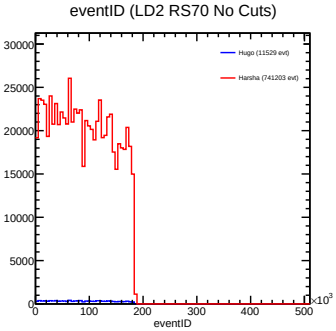




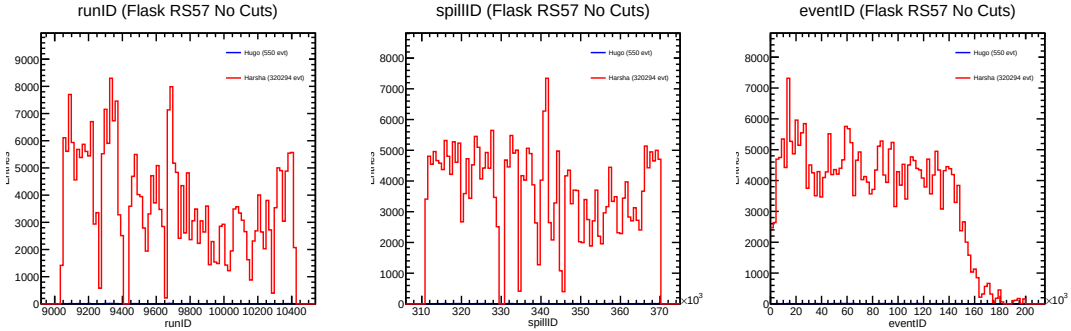
No Selection: LD2 Roadset 70





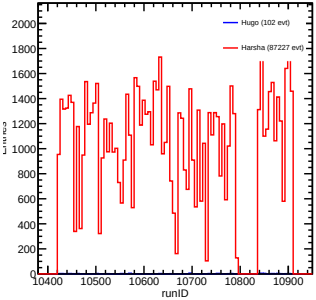


No Selection: Flask Roadset 57

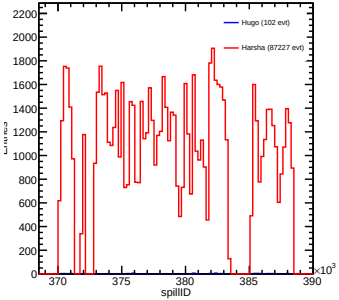


No Selection: Flask Roadset 59

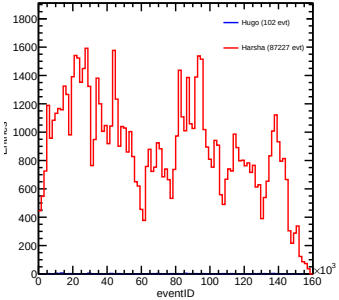
runID (Flask RS59 No Cuts)



spillID (Flask RS59 No Cuts)

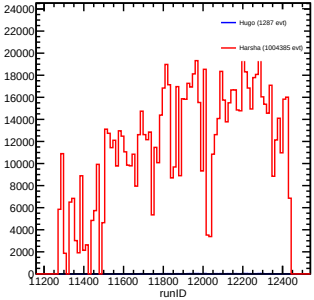


eventID (Flask RS59 No Cuts)

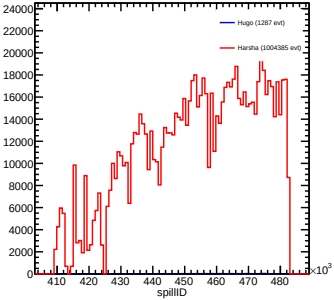


No Selection: Flask Roadset 62

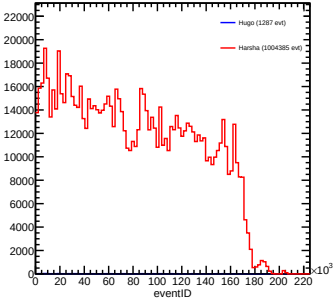
runID (Flask RS62 No Cuts)



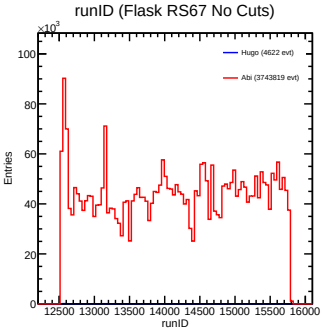
spillID (Flask RS62 No Cuts)

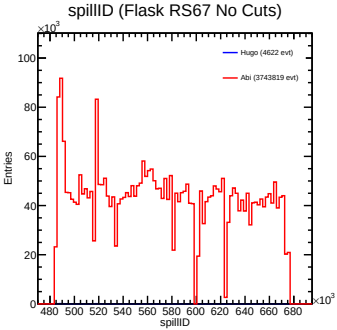


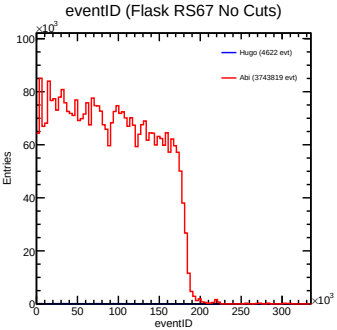
eventID (Flask RS62 No Cuts)



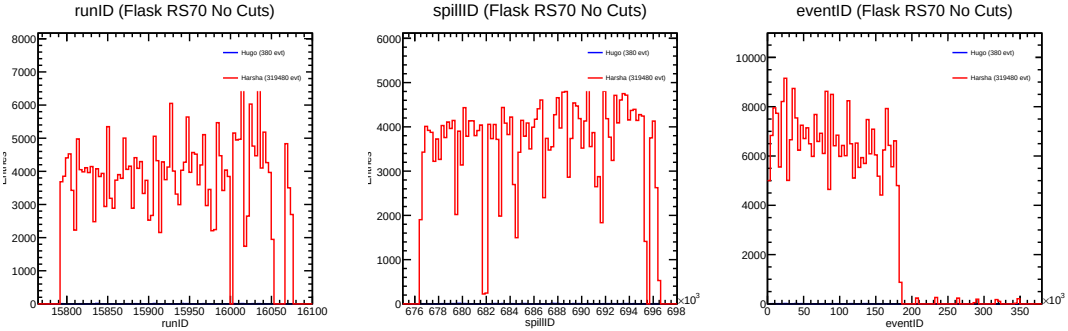
No Selection: Flask Roadset 67





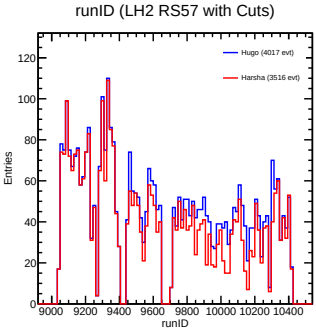


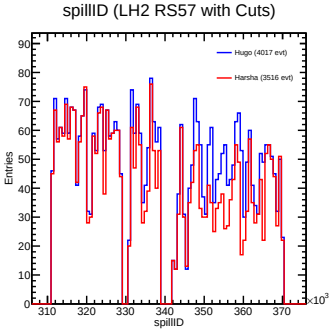
No Selection: Flask Roadset 70

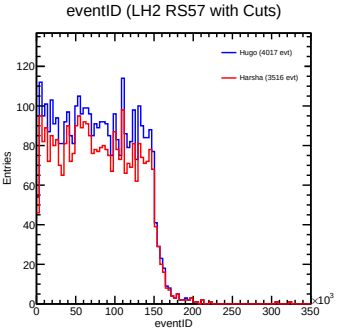


Plots With Event Selection (Chuck Cuts)

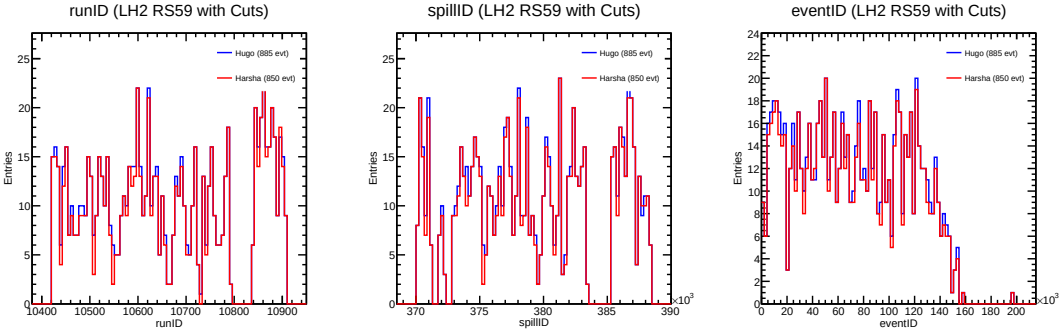
With Selection: LH2 Roadset 57



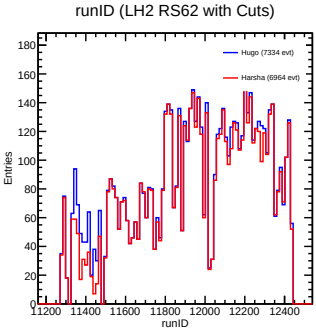


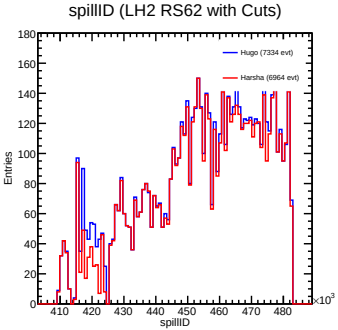


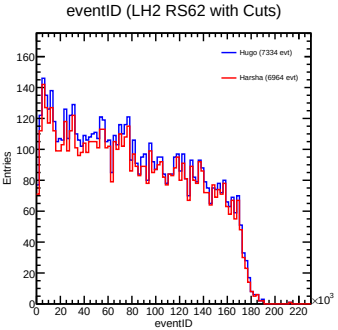
With Selection: LH2 Roadset 59



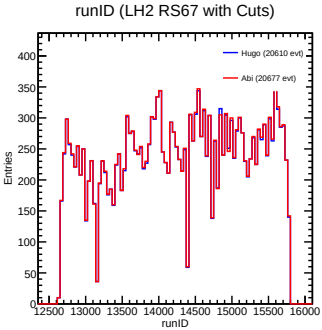
With Selection: LH2 Roadset 62

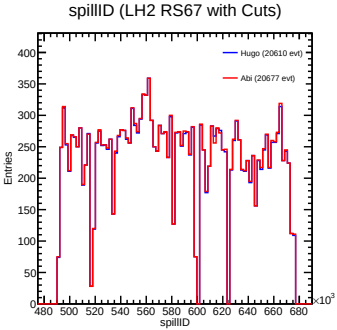


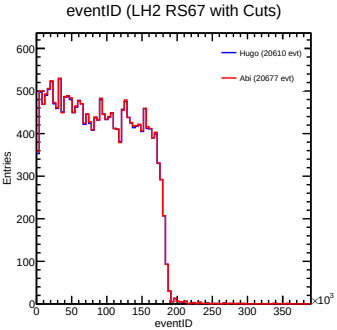




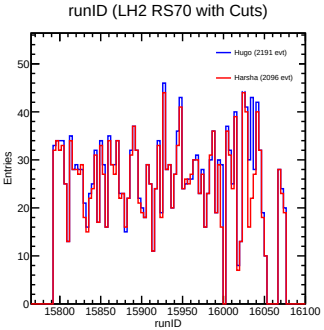
With Selection: LH2 Roadset 67

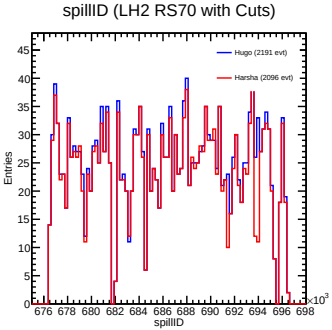


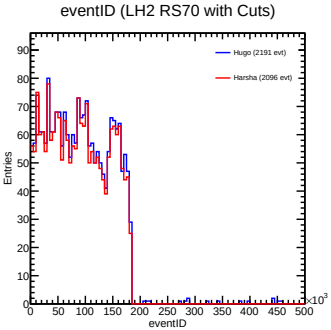




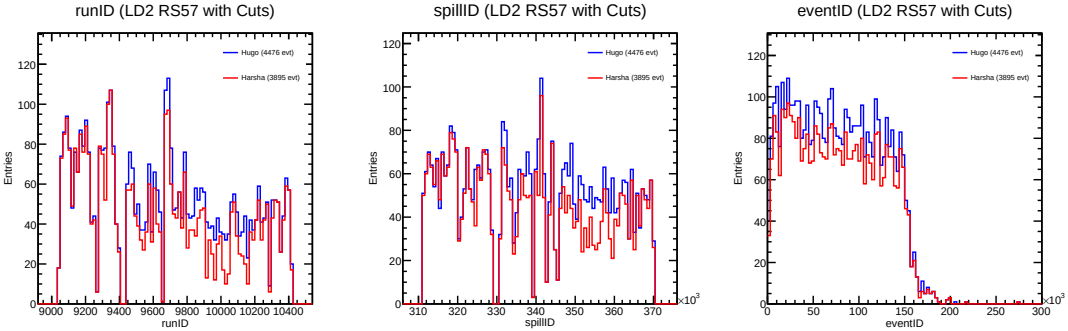
With Selection: LH2 Roadset 70



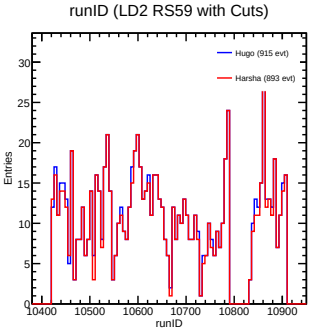


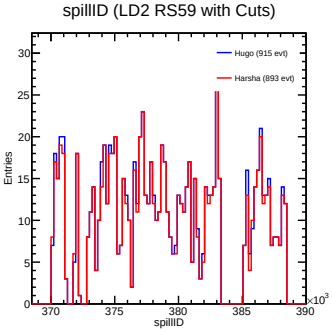


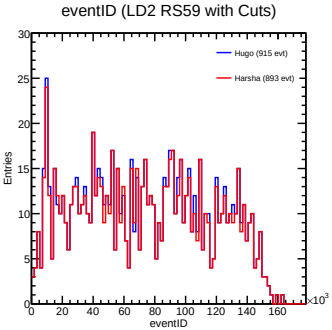
With Selection: LD2 Roadset 57



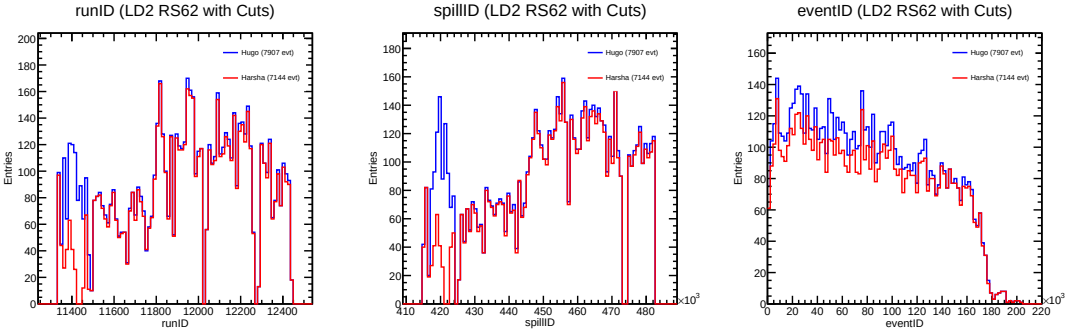
With Selection: LD2 Roadset 59



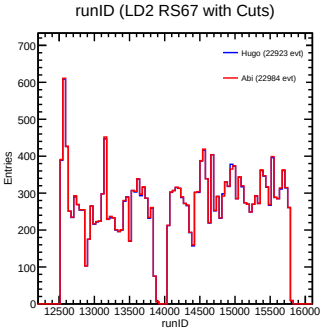


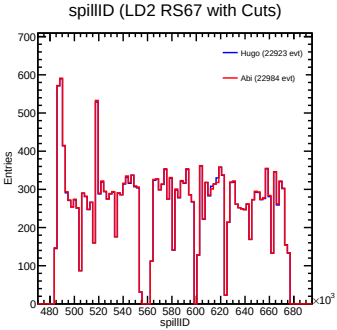


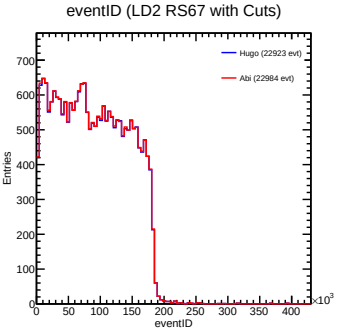
With Selection: LD2 Roadset 62



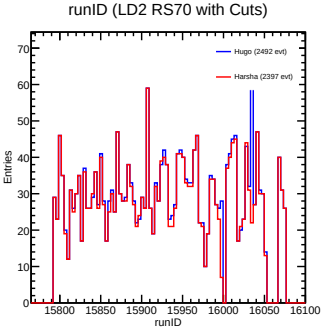
With Selection: LD2 Roadset 67

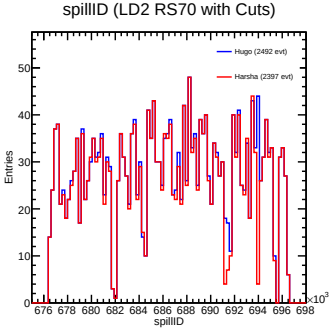


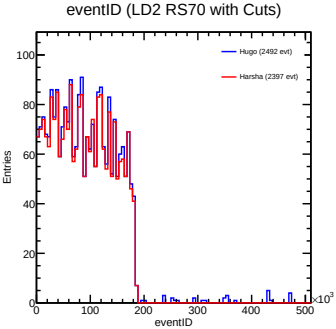




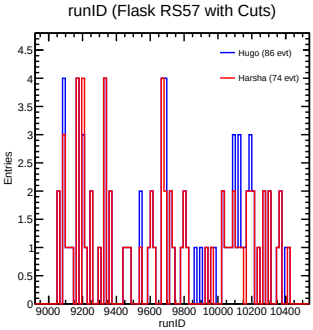
With Selection: LD2 Roadset 70

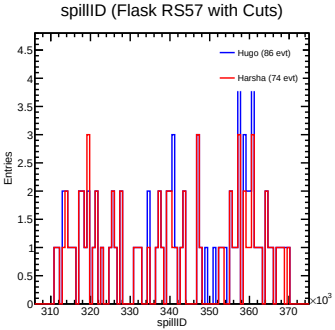


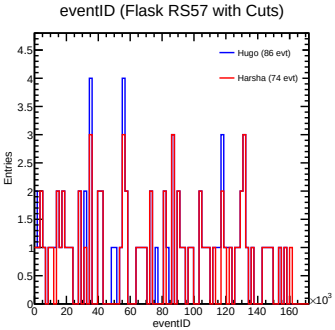




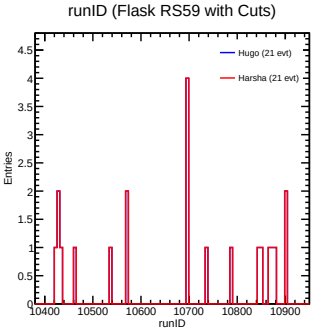
With Selection: Flask Roadset 57

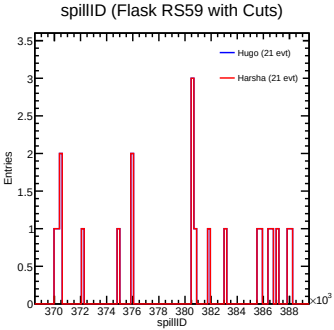


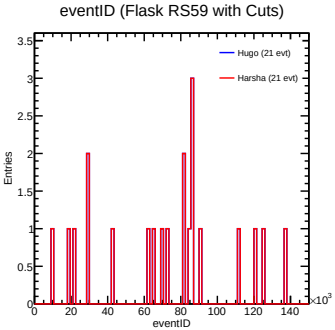




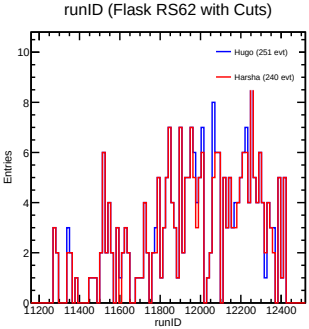
With Selection: Flask Roadset 59

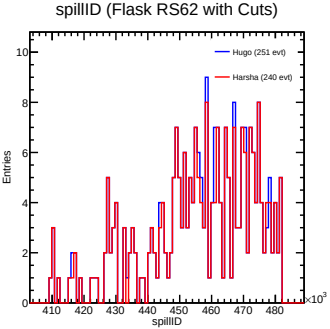


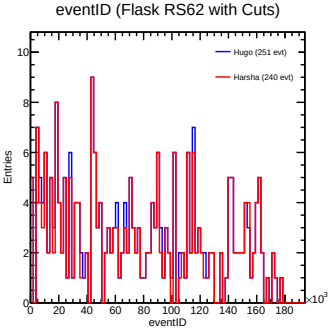




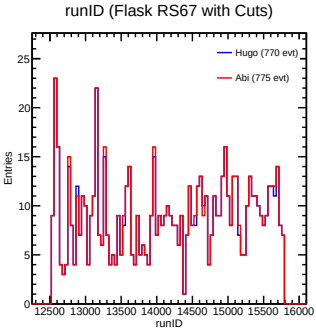
With Selection: Flask Roadset 62

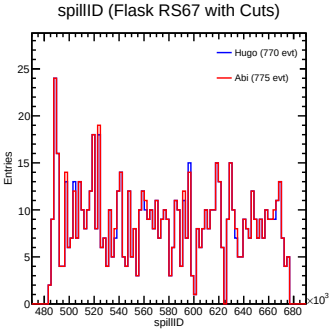


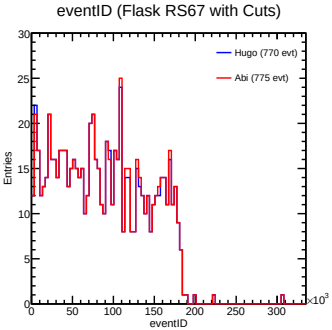
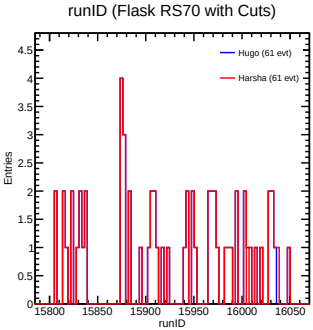
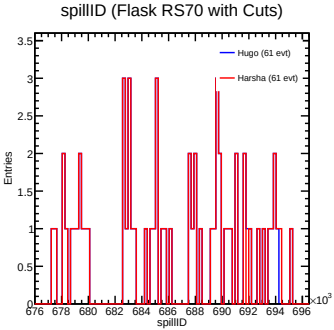
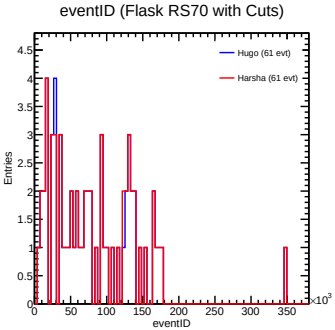




With Selection: Flask Roadset 67







With Selection: Flask Roadset 70